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DMF-Net: A Dual-domain Multi-scale 3D Fusion Framework for CT Synthesis in MRI-only Radiotherapy

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Abstract—Conventional radiotherapy relies on CT for electron density-based dosimetry. Despite that MRI offers superior soft-tissue contrast and avoids radiation exposure, the lack of electron density information in MRI limits its direct use in treatment planning. Synthetic CT offers a radiation-free alternative for generating the necessary electron density information, thus in principle enables the implementation of MRI-only radiotherapy. Most existing CT synthesis methods rely mainly on spatial domain representations, which limits their ability to capture high frequency details. In addition, conventional Vision Transformers use fixed attention windows, which weaken the modeling of long-range semantic relationships across complex anatomical structures. We propose a novel Dual-domain Multi-scale 3D Fusion framework (DMF-Net). Specifically, we first introduce a Multi-scale Perception Module (MPM) that leverages multi-scale token partitioning and adaptive window configurations to extend the receptive field and capture multi-directional structural contexts. Subsequently, we develop a Dual-domain Aggregation Module (DAM) that integrates frequency-domain features with spatial representations to mitigate spectral bias and preserve high-frequency textural details. Experimental results on brain and pelvis anatomical regions demonstrate that DMF-Net achieves state-of-the-art performance, which provides a reliable solution for MRI-based CT synthesis, advancing the clinical feasibility of MRI-only radiotherapy.

Index Terms—MR-to-CT synthesis, Transformer, Deep Learning, Radiotherapy, Multi scale

I. INTRODUCTION

Magnetic resonance imaging (MRI) has been increasingly integrated into modern radiotherapy treatment planning (RTP) due to its superior soft-tissue contrast, which facilitates more precise delineation of tumors, adjacent critical structures, and organs at risk (OARs) [1]. In routine clinical RTP, however, radiation dose calculation relies on patient-specific electron density information derived from the Hounsfield Units (HU) of computed tomography (CT) images. As a result, current clinical workflows typically require both MR and CT scans, with anatomical structures delineated on MR images being transferred to CT images via image registration [2]. This combined MR–CT-guided RTP is inherently susceptible to spatial misalignment and temporal anatomical changes between the two imaging acquisitions, potentially leading to dose calculation inaccuracies and compromised treatment precision [3]. Therefore, the synthesis of CT images directly from MR images has emerged as a clinically meaningful solution, enabling MR-only radiotherapy treatment planning and reducing registration uncertainty as well as additional radiation exposure [4] [5].

Deep learning-based approaches [6] [7] [8] [9] are widely used for MR-to-CT synthesis, with most recent studies adopting end-to-end learning frameworks. Typical methods [10] [11] [12] include CNN-based and Transformer-based architectures that learn voxel-wise intensity mapping while incorporating global contextual information. Recently, diffusion models [13] have also been explored for MR-to-CT synthesis due to their strong generative capability and high image quality, showing promising performance in preserving anatomical realism. However, diffusion-based methods [14] [15] usually rely on iterative denoising processes with a large number

of sampling steps, which leads to high computational cost and long inference time [16] [17], limiting their applicability in time-sensitive radiotherapy workflows, particularly for 3D volumetric synthesis.

Despite their structural diversities, most current methods are constrained by two shared limitations. First, methods [18] [19] predominantly rely on spatial-domain representations, which inherit an intrinsic spectral bias that prioritizes the learning of low-frequency components. This tendency typically leads to the oversmoothing of high-frequency details, resulting in the insufficient recovery of sharp bone boundaries and fine structural textures. Second, although Transformer-based models [20] [21] improve global context modeling, conventional Vision Transformers typically employ fixed-size attention windows, which restrict the effective receptive field and weaken the modeling of long-range and multi-directional semantic dependencies across complex anatomical structures. These limitations reduce structural consistency and anatomical fidelity in synthetic CT (sCT) images, which are critical for ensuring dosimetric accuracy in MRI-only radiotherapy.

In this paper, we propose a novel Dual-domain Multi-scale 3D Fusion Network (DMF-Net) for MR-to-CT synthesis, which is built upon a conditional generative adversarial learning framework. The proposed DMF-Net is designed to jointly exploit complementary information from both spatial and frequency domains while enhancing long-range contextual modeling in volumetric data. Specifically, we propose a Dual-domain Fusion Module that integrates frequency-domain features with spatial representations, enabling effective preservation of high-frequency textures and sharp structural details that are often degraded in spatial-domain-only learning. Furthermore, to overcome the restricted receptive field induced by fixed-window self-attention, we introduce a Multi-scale Window Self-Attention mechanism that employs multiple window sizes to adaptively capture multi-directional and long-range semantic dependencies across complex anatomical structures. By combining dual-domain feature fusion with multi-scale attention in a unified 3D framework, DMF-Net achieves more accurate and structurally consistent sCT generation. Our contributions can be summarized as follows:

- We propose a novel GAN-based dual-domain multi-scale 3D framework for MR-to-CT synthesis, which jointly exploits spatial and frequency-domain representations to mitigate spectral bias and enhance high-frequency detail preservation.
- We introduce a Multi-scale Perception Module (MPM) that overcomes the limitations of fixed-window attention and improves long-range and multi-directional anatomical modeling in 3D volumetric data, together with a Dual-domain Aggregation Module (DAM) that integrates frequency-domain features with spatial features.
- Extensive experiments on multi-region clinical datasets demonstrate that the proposed DMF-Net achieves superior performance.

II. METHOD

The overall architecture of the proposed DMF-Net for 3D MR-to-CT synthesis is illustrated in Fig. 1. Our framework is based on a conditional Generative Adversarial Network (cGAN). It consists of a 3D generator and a 3D discriminator. The generator is designed to learn the complex mapping from 3D MRI volumes to sCT images. To improve the synthesis quality, the generator incorporates two key components in sequence. First, a Multi-scale Window Transformer captures global contextual information through multi-resolution patch analysis. Second, a Dual-domain Aggregation Module (DAM) integrates spatial and frequency information to refine anatomical details. The 3D discriminator aims to distinguish real CT volumes from sCT volumes conditioned on the input MRI, thereby encouraging the generator to produce anatomically consistent and realistic CT images.

A. Multi-scale Perception Module

To address the limited receptive fields of traditional CNNs and capture long-range dependencies, we introduce the Multi-scale Perception Module. This module enables the generator to extract global spatial information and multi-scale contextual features across complex anatomical structures. It is integrated into each downsampling branch of the encoder to enhance feature representation. Specifically, MPM uses 3D convolutions with various kernel sizes to partition the input into non-overlapping multi-scale tokens to perceive features at different resolutions. For an input MR volume $x_{MR} \in \mathbb{R}^{H \times W \times D \times C}$, the token representations at the s -th scale can be defined as:

$$T_s = \text{Reshape}(\text{Conv}_{3D}(x_{MR}; k_s)), \quad (1)$$

where $\text{Conv}_{3D}(\cdot; k_s)$ denotes a 3D convolution with kernel size k_s , and $\text{Reshape}(\cdot)$ flattens the resulting feature maps into a sequence of 3D tokens.

These tokens are then input the Multi-scale Window-based Transformer (MSwin-T) [10], which can be expressed as:

$$Z_s = \text{MSwin-T}(T_s), \quad (2)$$

where self-attention mechanism can capture multi-directional semantic relationships.

Finally, the token-aware information is fused with the local features from the CNN encoder:

$$\begin{aligned} M_0 &= \phi_0(x_{MR}), \\ M_s &= Z_s \odot \phi_s(M_{s-1}), \end{aligned} \quad (3)$$

where $\phi_s(\cdot)$ denotes the s -th convolutional layer of the CNN-based encoder, and $\odot$ denotes a 1×1 convolution-based aggregation operation. This design guarantees a more effective integration of global context and local details, leading to improved structural consistency in the sCT images.

B. Dual-domain Aggregation Module

While spatial-domain features excel at representing anatomical structures, they often fail to capture fine textures. To mitigate spectral bias and recover high-frequency details, we

propose the Dual-domain Aggregation Module. DAM addresses this by jointly aggregating features from both the spatial and frequency domains. As illustrated in the "Dual-domain Aware" section of Fig. 1, DAM first transforms the spatial feature map F_s into the frequency domain using the Fast Fourier Transform (FFT):

$$F_f = \mathcal{F}(F_s), \quad (4)$$

where $\mathcal{F}$ denotes the FFT operation, and F_f represents the corresponding frequency domain features. Both F_s and F_f are independently feed in MSwin-T blocks to capture domain-specific global dependencies. The enhanced frequency-domain features are then converted back to the spatial representation via the inverse FFT:

$$\begin{aligned} \hat{F}_s &= \text{MSwin-T}(F_s), \\ \hat{F}_f &= \mathcal{F}^{-1}(\text{MSwin-T}(F_f)), \end{aligned} \quad (5)$$

where $\mathcal{F}^{-1}$ denotes the inverse FFT operation. The Fusion block serves as the core component of DAM and is responsible for adaptively integrating the spatial-domain features $\hat{F}_s$ and the frequency-domain $\hat{F}_f$ features. Specifically, $\hat{F}_s$ and $\hat{F}_f$ are first concatenated along the channel dimension and recalibrated by a Channel-Attention mechanism, enabling the network to emphasize informative channels across domains. To further capture structural information at different receptive fields, the refined features are then processed by a multi-scale branch consisting of parallel 3D convolutions with kernel sizes $3 \times 3 \times 3$, $5 \times 5 \times 5$ and $7 \times 7 \times 7$. These multi-scale features are adaptively fused through an attention-based weighting strategy implemented via a Softmax function, allowing the model to selectively aggregate complementary information from different scales. Finally, the fused representation is combined with the original spatial feature to produce the aggregated output feature F_{out} , which can be defined as:

$$F_{out} = \text{Fusion}(\hat{F}_s, \hat{F}_f) \quad (6)$$

DAM guarantees the preservation of both global anatomical consistency and high-frequency textural realism, significantly enhancing the fidelity of the synthetic CT images.

C. Loss Function

We employ a multi-component loss function to optimize DMF-Net to ensure both voxel-level accuracy and generative realism. The total objective function consists of three components: an adversarial loss, an L_1 distance loss, and a VGG-based perceptual loss.

To ensure that the synthetic CT images follow the distribution of real clinical data, we employ an adversarial loss L_{adv} . This loss drives a min-max game between the generator G and the discriminator D . It encourages the generator to produce high-fidelity images that are indistinguishable from ground truth CT. We use a 3D PatchGAN [22] as the discriminator. The adversarial loss is defined as:

$$\begin{aligned} L_{adv} &= \mathbb{E}_{x_{MR}, x_{CT}} [\log D(x_{MR}, x_{CT})] \\ &\quad + \mathbb{E}_{x_{MR}} [\log (1 - D(x_{MR}, G(x_{MR})))]. \end{aligned} \quad (7)$$

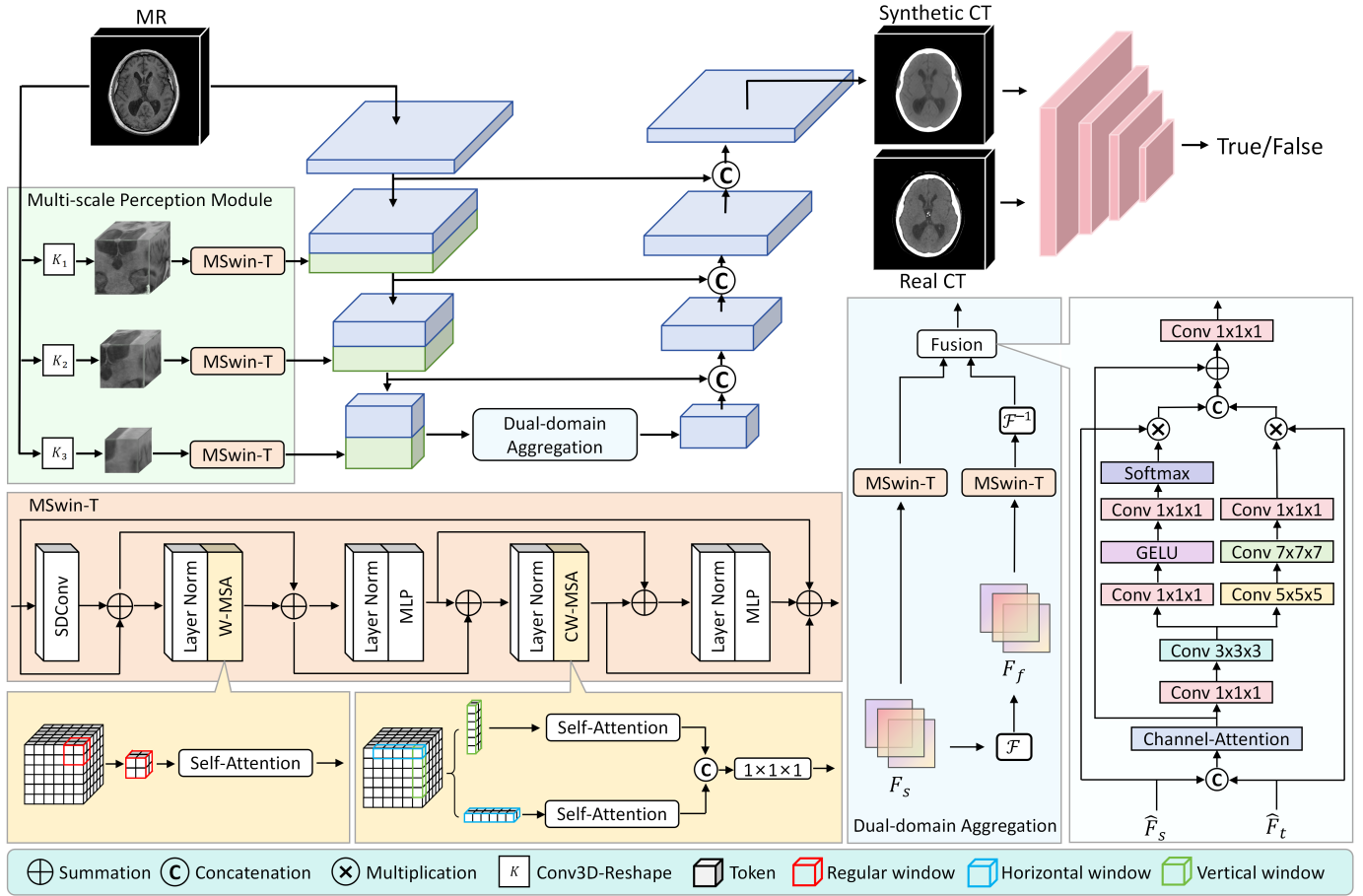


Fig. 1. Overview of the proposed DMF-Net.

where x_{MR} and x_{CT} represent the input MRI and the corresponding real CT volume.

To minimize intensity differences and ensure accurate HU estimation, we implement the pixel-wise L_1 distance loss L_{L1} . This loss provides direct voxel-level supervision by measuring the absolute difference between the synthetic and real CT images. It is defined as:

$$L_{L1} = \|G(x_{MR}) - x_{CT}\|_1 \quad (8)$$

This objective ensures that the global anatomical structures and density values are correctly reconstructed.

To further enhance the capture of fine textures and structural details, we incorporate a VGG-based perceptual loss L_{VGG} . Unlike pixel-wise losses, this term compares high-level feature maps extracted by a pre-trained VGG-19 network ϕ [23]. It is defined as:

$$L_{VGG} = \|\phi(G(x_{MR})) - \phi(x_{CT})\|_1 \quad (9)$$

By leveraging semantic features, the model can better recover sharp bone boundaries and complex tissue interfaces that are critical for dosimetry.

Finally, the total loss function used to optimize the overall performance of the model is a weighted combination of the three components:

$$L_{total} = L_{adv} + \lambda_{L1} L_{L1} + L_{VGG} \quad (10)$$

where λ_{L1} is a hyper-parameter that balances generative realism with intensity accuracy. This comprehensive loss strategy enables DMF-Net to generate synthetic CT images with high structural fidelity and realistic intensity distributions.

III. EXPERIMENTS AND RESULTS

A. Dataset

A total of 193 paired CT-MR scans were retrospectively collected from patients at the Mayo Clinic in Arizona, including two anatomical regions: the brain (102 cases) and the pelvis (91 cases). For the brain region, MR scans were acquired using T2-weighted sequences that had an in-plane resolution of 0.60–1.27 mm, a slice thickness ranging from 0.6 to 4.0 mm, and comprised 140–357 slices per acquired volume. The paired CT scans were collected with in-plane spacing of 0.78–1.27 mm, slice thickness of 1.0 to 2.0 mm, and 199–547 slices per volume. For the pelvis region, MR images were acquired with T1-weighted sequences. The in-plane pixel spacing ranged from 0.59 to 1.17 mm, the slice thickness

ranged from 1.2 to 4.4 mm, and the number of slices per volume ranged from 30 to 280. The corresponding CT images were obtained with an in-plane spacing of 1.27 mm, a slice thickness of 2.0 mm, and 218-310 slices per volume. For each patient, the time interval between CT and MR scans was less than 7 days to reduce anatomical differences and ensure spatial consistency. All protected health information was removed before analysis. This retrospective study was approved by the Institutional Review Board of the Mayo Clinic in Arizona (IRB, 13-005709). 102 brain subjects were randomly divided into a training set of 81 subjects, a validation set of 5 subjects, and a testing set of 16 subjects. 91 pelvic subjects were randomly divided into a training set of 72 subjects, a validation set of 4 subjects, and a testing set of 15 subjects.

For data preprocessing, we first applied N4 Bias Field Correction [24] to all MR images to remove intensity inhomogeneities. The images were then resampled to a resolution of $1.0 \times 1.0 \times 1.0 \text{ mm}^3$ for brain data and $1.0 \times 1.0 \times 2.5 \text{ mm}^3$ for pelvis data. To eliminate non-anatomical noise, the CT couch was removed from all CT scans. Each patient's CT image was aligned to the corresponding MR image using rigid registration [25]. Following the registration process, the brain volumes were cropped to preserve cranial anatomy. Both MR and CT images were uniformly normalized to an intensity range of $[-1, 1]$. Finally, 3D patches with a size of $128 \times 128 \times 8$ were extracted for model training and evaluation.

B. Evaluation Metrics

To quantitatively evaluate the quality of the synthetic CT images, we employed three widely accepted metrics, including the Mean Absolute Error (MAE), Structural Similarity Index Measure (SSIM), and the Peak Signal-to-Noise Ratio (PSNR). MAE measures the voxel-wise intensity accuracy in HU. SSIM evaluates the preservation of structural information, including perceived luminance, contrast, and anatomical boundaries. PSNR is used to assess the signal fidelity and noise levels of the generated images. Lower MAE, higher SSIM and PSNR values indicate better synthesis performance.

C. Implementation Details

The proposed DMF-Net was implemented in Python using the PyTorch library. We utilized the Adam optimizer with momentum parameters $\beta_1 = 0.5$ and $\beta_2 = 0.999$. We trained DMF-Net for 200 epochs with an initial learning rate of 0.0002 that was dynamically adjusted using a "poly" decay strategy. We set the batch size to 4 and assigned the trade-off hyperparameter $\lambda_{L1} = 20$ as established in the total loss function. All experiments were conducted on a single NVIDIA RTX 4070 super GPU with 12GB of memory to ensure consistent computational conditions.

D. Results

The quantitative results are summarized in Table I, which shows that DMF-Net achieves the lowest MAE, and highest SSIM and PSNR on both brain and pelvis region. Specifically, For the brain region, DMF-Net achieves an MAE of $27.23 \pm$

4.39 HU , an SSIM of 0.921 ± 0.132 , and a PSNR of $28.63 \pm 0.83 \text{ dB}$. For the pelvis region, it reaches an MAE of $36.66 \pm 3.29 \text{ HU}$, an SSIM of 0.846 ± 0.013 , and a PSNR of $29.46 \pm 1.02 \text{ dB}$. These results indicate that our framework can accurately estimate HU values, providing a reliable basis for radiotherapy dose calculation. This can be attributed to two main reasons. First, the Multi-scale Perception Module enables the generator to capture long-range dependencies and global spatial information, ensuring anatomical consistency. Second, the Dual-domain Aggregation Module mitigates spectral bias by integrating frequency-domain features with spatial representations. This dual-path refinement allows for the precise reconstruction of high-frequency details, such as sharp bone boundaries and complex soft tissue textures. Notably, the low standard deviation across subjects demonstrates the stability of DMF-Net in handling diverse clinical cases. This stability is largely due to our multi-scale window attention mechanism, which adaptively processes varying anatomical geometries and organ sizes. By leveraging multiple window scales, the network maintains a robust receptive field that is less sensitive to individual anatomical differences. Furthermore, the dual-domain fusion strategy synchronizes spatial structures with frequency details, which effectively prevents significant intensity fluctuations across different imaging sequences. This ensures that DMF-Net provides consistent and reliable synthetic CT generation, advancing its feasibility for clinical MRI-only radiotherapy.

To evaluate the performance of the proposed DMF-Net, we compared it with other state-of-the-art methods, including SwinUNETR [26], 3D-Unet [27], and MTT-Net [10]. Swin UNETR [26] employs a hierarchical Swin Transformer with shifted-window self-attention and self-supervised pre-training to model long-range spatial dependencies, but it is not specifically designed for cross-modality synthesis and lacks explicit intensity and frequency-domain constraints, leading to suboptimal recovery of high-frequency CT details. MTT-Net [10] models multi-region and multi-sequence MR-to-CT synthesis by leveraging multi-scale token representations within a unified transformer framework; however, its interactions are confined to the spatial domain without frequency-aware supervision, which limits the preservation of sharp anatomical boundaries. In contrast, our method explicitly integrates dual-domain aggregation and adaptive multi-scale perception, enabling more accurate structural preservation and consistently superior quantitative performance.

E. Ablation Study

To verify the individual contributions of MPM and DAM, we conducted four ablation experiments, including (1) Baseline (Pix2pix with MSwin-T), (2) Baseline + MPM, (3) Baseline + DAM, and (4) DMF-Net (Baseline + MPM + DAM). The quantitative results are summarized in Table II.

As shown in Table II, the inclusion of each module lead to incremental improvements in synthesis performance. "Baseline + DDA" improves the PSNR to $27.67 \pm 0.70 \text{ dB}$, proving that dual-domain features help recover high-frequency details.

TABLE I

PERFORMANCE COMPARISON BETWEEN OUR PROPOSED AND OTHER DIFFERENT METHODS ON BRAIN DATA AND PELVIS DATA. THE NUMBER ARE IN MEAN $\pm$ STANDARD DEVIATION. BOLD VALUES INDICATE THE BEST RESULTS FOR EACH COLUMN.

Methods	Brain				Pelvis			
	MAE (HU) ↓	SSIM ↑	PSNR (dB) ↑	<i>p</i> -value	MAE (HU) ↓	SSIM ↑	PSNR (dB) ↑	<i>p</i> -value
SwinUNETR [26]	52.61 $\pm$ 11.56	0.890 $\pm$ 0.026	28.65$\pm$1.49	<1e-7	54.68 $\pm$ 8.07	0.836 $\pm$ 0.022	27.73 $\pm$ 1.28	<1e-9
3D-Unet [27]	37.26 $\pm$ 8.56	0.892 $\pm$ 0.018	26.87 $\pm$ 1.05	<1e-6	55.47 $\pm$ 2.41	0.767 $\pm$ 0.026	29.15 $\pm$ 0.53	<1e-10
MTT-Net [10]	30.28 $\pm$ 4.51	0.908 $\pm$ 0.013	27.99 $\pm$ 0.71	<1e-5	46.91 $\pm$ 2.47	0.739 $\pm$ 0.021	28.30 $\pm$ 0.94	<1e-11
DMF-Net	27.23$\pm$4.39	0.921$\pm$0.132	28.63 $\pm$ 0.83	-	36.66$\pm$3.29	0.846$\pm$0.013	29.46$\pm$1.02	-

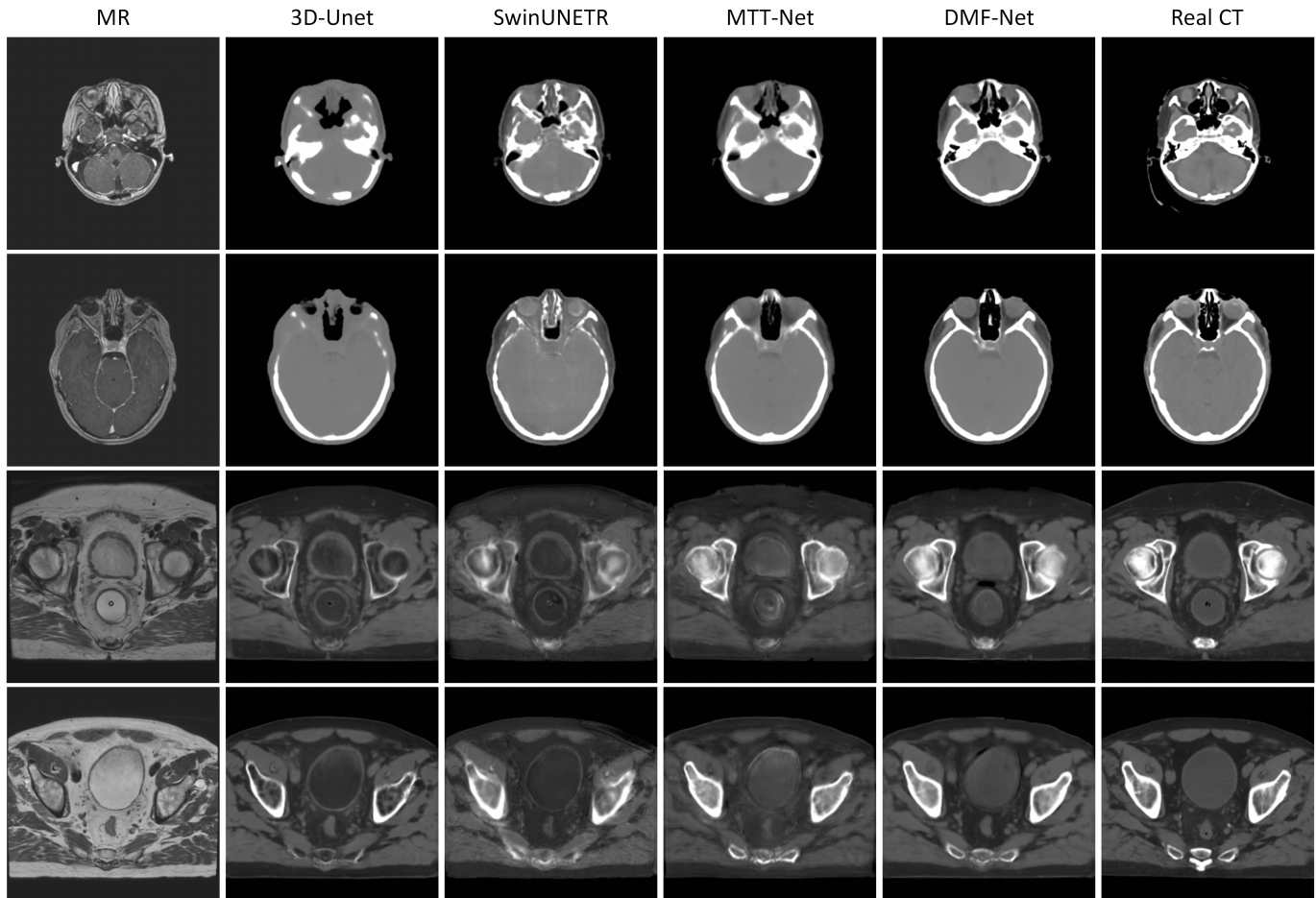


Fig. 2. Visualization of the synthesis results of different methods on brain data and pelvis data.

Similarly, "Baseline + MPM" achieves an MAE of 32.00 ± 5.54 HU, demonstrating that capturing multi-scale global context is essential for reducing structural errors. Finally, the complete DMF-Net achieves the best performance across all metrics, with an MAE of 27.23 ± 4.39 HU for brain data and 36.66 ± 3.29 HU for pelvis data. These results confirm that the synergy between global spatial modeling and dual-domain refinement is critical for high-fidelity CT synthesis.

Fig. 3 shows the visual results and corresponding difference maps. The difference maps represent the voxel-wise absolute error between the synthetic CT and the ground truth. Compared to the baseline, "Baseline + MPM" reduces errors in

large anatomical structures by ensuring better global consistency. "Baseline + DDA" significantly minimizes intensity discrepancies at bone-tissue interfaces, as shown by the reduced brightness in the difference maps at the skull and pelvic boundaries. The full DMF-Net generates the most accurate synthetic CT with the smallest error residuals. This qualitative evidence highlights the effectiveness of our proposed modules in maintaining anatomical fidelity and reducing artifacts.

IV. CONCLUSION

In this study, we proposed DMF-Net, a dual-domain multi-scale 3D framework for MR-to-CT synthesis in MRI-only

TABLE II

EXPERIMENTAL RESULTS OF THE ABLATION STUDY ON BRAIN DATA AND PELVIS DATA. THE NUMBER ARE IN MEAN±STANDARD DEVIATION. BOLD VALUES INDICATE THE BEST RESULTS FOR EACH COLUMN.

Methods	Brain				Pelvis			
	MAE (HU) ↓	SSIM ↑	PSNR (dB) ↑	p-value	MAE (HU) ↓	SSIM ↑	PSNR (dB) ↑	p-value
Baseline	35.75±5.98	0.891±0.018	26.77±0.85	<1e-9	47.94±2.90	0.775±0.160	28.12±0.60	<1e-11
Baseline+DDA	31.57±4.52	0.901±0.014	27.67±0.70	<1e-6	45.48±2.33	0.780±0.019	28.50±0.69	<1e-8
Baseline+MPM	32.00±5.54	0.906±0.017	27.51±0.88	<1e-6	43.52±2.83	0.804±0.015	28.70±0.75	<1e-9
DMF-Net	27.23±4.39	0.921±0.132	28.63±0.83	-	36.66±3.29	0.846±0.013	29.46±1.02	-

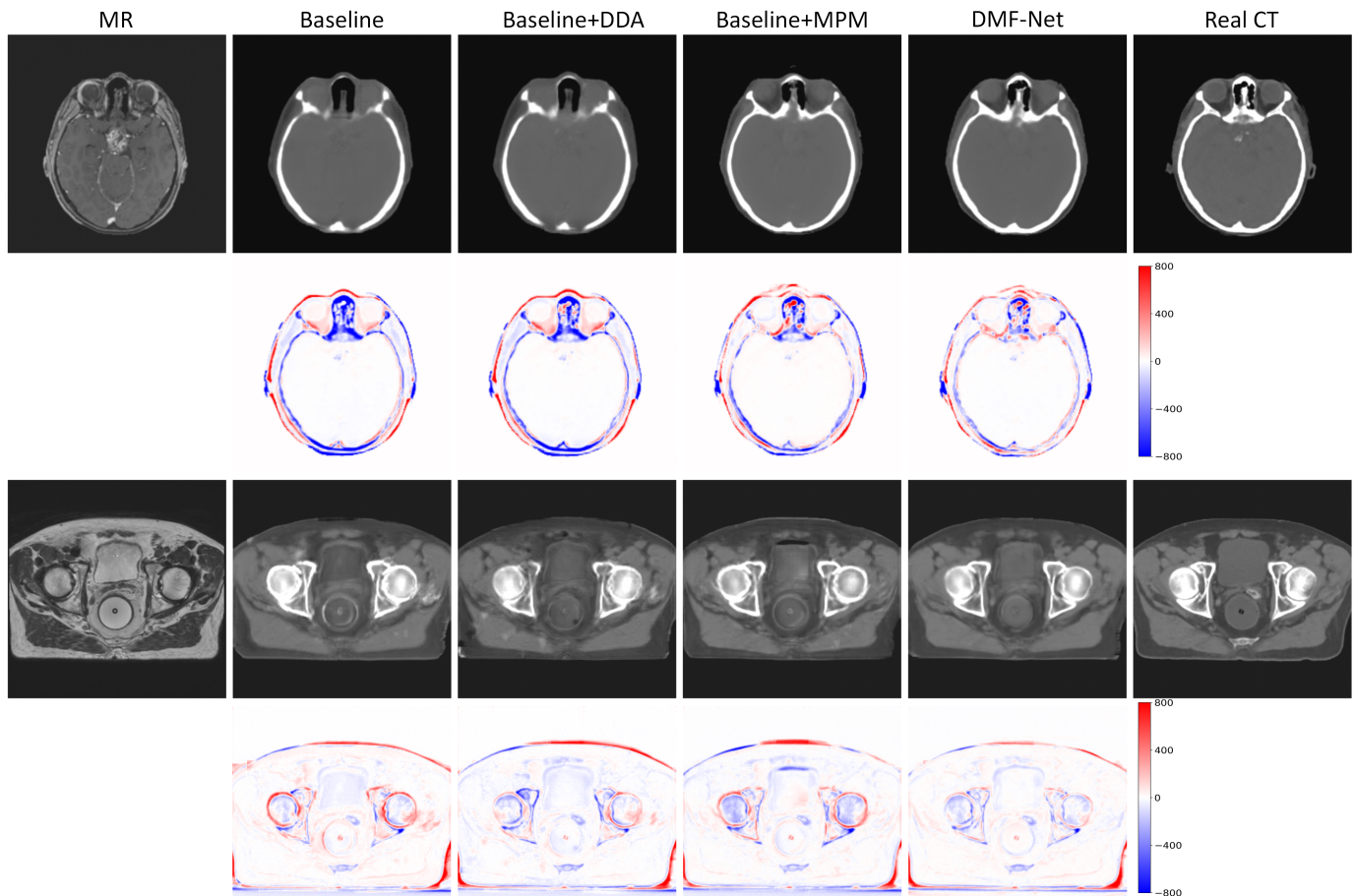


Fig. 3. Visual comparison of representative output and difference map results produced by DMF-Net and its variants on brain data and pelvis data.

radiotherapy. DMF-Net employs a Multi-scale Perception Module to capture global anatomical context and a Dual-domain Aggregation Module to preserve high-frequency details. Experiments on multi-center brain and pelvis datasets demonstrate that DMF-Net achieves superior image quality and dosimetric accuracy. The proposed DMF-Net provides reliable electron density maps for MRI-only radiotherapy, thereby supporting optimized treatment workflows. Future work will validate the framework across additional centers with varied imaging protocols and evaluate the dosimetric reliability of sCT in real-world treatment planning.

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REFERENCES

- [1] G. Whitfield, S. Kennedy, I. Djoukhar, and A. Jackson, "Imaging and target volume delineation in glioma," *Clinical Oncology*, vol. 26, no. 7, pp. 364–376, 2014, central nervous system tumours.
- [2] A. Fiorentino, R. Caivano, P. Pedicini, and V. Fusco, "Clinical target volume definition for glioblastoma radiotherapy planning: magnetic resonance imaging and computed tomography," *Clinical and Translational Oncology*, vol. 15, no. 9, pp. 754–758, 2013.
- [3] M. K. Kalra, M. M. Maher, T. L. Toth, L. M. Hamberg, M. A. Blake, J.-A. Shepard, and S. Saini, "Strategies for ct radiation dose optimization," *Radiology*, vol. 230, no. 3, pp. 619–628, 2004.
- [4] J. S. Witt, S. A. Rosenberg, and M. F. Bassetti, "Mri-guided adaptive radiotherapy for liver tumours: visualising the future," *The Lancet Oncology*, vol. 21, no. 2, pp. e74–e82, 2020.
- [5] R. Otazo, P. Lambin, J.-P. Pignol, M. E. Ladd, H.-P. Schlemmer, M. Baumann, and H. Hricak, "Mri-guided radiation therapy: an emerging paradigm in adaptive radiation oncology," *Radiology*, vol. 298, no. 2, pp. 248–260, 2021.
- [6] J. Zheng, Z. Shen, L. Zhang, and Q. Chen, "Structure-guided mr-to-ct synthesis with spatial and semantic alignments for attenuation correction of whole-body pet/mr imaging," *Medical Image Analysis*, vol. 103, p. 103622, 2025.
- [7] R. Wang, A. F. Heimann, M. Tannast, and G. Zheng, "Cyclesgan: A cycle-consistent and semantics-preserving generative adversarial network for unpaired mr-to-ct image synthesis," *Computerized Medical Imaging and Graphics*, vol. 117, p. 102431, 2024.
- [8] L. Zhong, Z. Chen, H. Shu, Y. Zheng, Y. Zhang, Y. Wu, Q. Feng, Y. Li, and W. Yang, "Qacl: Quartet attention aware closed-loop learning for abdominal mr-to-ct synthesis via simultaneous registration," *Medical image analysis*, vol. 83, p. 102692, 2023.
- [9] Y. Peng, J. Sun, Y. Ren, D. Li, and Y. Guo, "A histogram-driven generative adversarial network for brain mri to ct synthesis," *Knowledge-Based Systems*, vol. 277, p. 110802, 2023.
- [10] L. Zhong, Z. Chen, H. Shu, K. Zheng, Y. Li, W. Chen, Y. Wu, J. Ma, Q. Feng, and W. Yang, "Multi-scale tokens-aware transformer network for multi-region and multi-sequence mr-to-ct synthesis in a single model," *IEEE Transactions on Medical Imaging*, vol. 43, no. 2, pp. 794–806, 2024.
- [11] B. Zhao, T. Cheng, X. Zhang, J. Wang, H. Zhu, R. Zhao, D. Li, Z. Zhang, and G. Yu, "Ct synthesis from mr in the pelvic area using residual transformer conditional gan," *Computerized medical imaging and graphics*, vol. 103, p. 102150, 2023.
- [12] V. M. H. Phan, Y. Xie, B. Zhang, Y. Qi, Z. Liao, A. Perperidis, S. L. Phung, J. W. Verjans, and M.-S. To, "Structural attention: Rethinking transformer for unpaired medical image synthesis," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2024, pp. 690–700.
- [13] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," *Advances in neural information processing systems*, vol. 33, pp. 6840–6851, 2020.
- [14] C. Gong, Z. Huang, Z. Wu, M. Zhao, F. Xu, X. Bai, M. Feng, A. Granados, G. Meng, Z. Lei, and H. Liu, "Mr-to-ct translation using frequency-separated diffusion models," in *2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI)*, 2025, pp. 1–5.
- [15] A. Tapp, A. Parida, C. Zhao, V. Lam, N. Lepore, S. M. Anwar, and M. G. Linguraru, "Mr to ct synthesis using 3d latent diffusion," in *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024, pp. 1–5.
- [16] Q. Gao, Z. Li, J. Zhang, Y. Zhang, and H. Shan, "Corediff: Contextual error-modulated generalized diffusion model for low-dose ct denoising and generalization," *IEEE Transactions on Medical Imaging*, vol. 43, no. 2, pp. 745–759, 2023.
- [17] M. Li, J. Lin, C. Meng, S. Ermon, S. Han, and J.-Y. Zhu, "Efficient spatially sparse inference for conditional gans and diffusion models," *Advances in neural information processing systems*, vol. 35, pp. 28 858–28 873, 2022.
- [18] X. Gu, Y. Zhang, W. Zeng, S. Zhong, H. Wang, D. Liang, Z. Li, and Z. Hu, "Cross-modality image translation: Ct image synthesis of mr brain images using multi generative network with perceptual supervision," *Computer Methods and Programs in Biomedicine*, vol. 237, p. 107571, 2023.
- [19] C. Lee, Y. H. Yoon, J. Sung, J. W. Kim, Y. Cho, J. Kim, J. Chun, and J. S. Kim, "Abdominal synthetic ct generation for mr-only radiotherapy using structure-conserving loss and transformer-based cycle-gan," *Frontiers in Oncology*, vol. 14, p. 1478148, 2025.
- [20] X. Zhu and Y. Li, "A latent multi-scale residual transformer approach for cross-modal medical image synthesis," *IEEE Access*, 2025.
- [21] Y. Li, S. Xu, Y. Lu, and Z. Qi, "Ct synthesis from mri with an improved multi-scale learning network," *Frontiers in Physics*, vol. 11, p. 1088899, 2023.
- [22] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 1125–1134.
- [23] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *arXiv preprint arXiv:1409.1556*, 2014.
- [24] N. J. Tustison, B. B. Avants, P. A. Cook, Y. Zheng, A. Egan, P. A. Yushkevich, and J. C. Gee, "N4itk: improved n3 bias correction," *IEEE transactions on medical imaging*, vol. 29, no. 6, pp. 1310–1320, 2010.
- [25] M. Jenkinson, C. F. Beckmann, T. E. Behrens, M. W. Woolrich, and S. M. Smith, "Fsl," *Neuroimage*, vol. 62, no. 2, pp. 782–790, 2012.
- [26] Y. Tang, D. Yang, W. Li, H. R. Roth, B. Landman, D. Xu, V. Nath, and A. Hatamizadeh, "Self-supervised pre-training of swin transformers for 3d medical image analysis," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 20 730–20 740.
- [27] A. Wolny, L. Cerrone, A. Vijayan, R. Tofanelli, A. V. Barro, M. Louveaux, C. Wenzl, S. Strauss, D. Wilson-Sánchez, R. Lymbouridou *et al.*, "Accurate and versatile 3d segmentation of plant tissues at cellular resolution," *Elife*, vol. 9, p. e57613, 2020.